MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestone 1 Proposal

AI-Based Student Dropout Early Warning System for Higher Education

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1. Executive Summary / Abstract

A practical and financially expensive challenge in institutions of higher education are student dropouts due to potential impacts on retention rates, educational outcomes, advising workloads, and resource efficiency regarding student academic support. This project plans to create an artificial intelligence (AI)-based early detection system with the goal of predicting whether a student has a high probability of dropping out from their institution of enrollment, remaining enrolled, or graduating. The primary dataset planned for use in this project is the UCI Predict Students' Dropout and Academic Success dataset. Machine Learning models including Logistic Regression, Decision Tree, Random Forest, and XGBoost will be employed in this project. Additionally, explainable AI techniques specifically SHAP will be utilized to demonstrate which input feature(s) provide the greatest amount of contribution towards each model's predictions. Ultimately, the expected result of this project is not a complete production-ready platform; however, a simple prototype of a decision-making process workflow and dashboard intended to assist educators in identifying students who have greater risk profiles earlier than would normally occur without the aid of this system.

2. Problem Statement & Project Objectives

Formal Problem Statement

Develop an early warning system based on machine learning for the purpose of predicting a student's dropout risk utilizing publically available student data. Use this to determine probable student outcomes (Dropout, Enrolled, and Graduate). The proposed system shall be both interpretable by its users, feasible for implementation prior to the end of an academic term, and assist users in developing meaningful early intervention plans. All development will utilize commonly used software packages (Python, Scikit-learn), which can be run locally on laptops or via Google Colab..

SMART Analysis

The project is unique because it is focused on one specific problem: identifying those students who have some level of risk of being a "dropout" before the situation has become difficult to rectify. It does not attempt to find solutions to all issues associated with student success. Rather, it focuses on a manageable prediction problem utilizing structured student record information and explainable machine learning methods.

The project is quantifiable because the performance of each developed model can be measured through use of standard classification metrics such as accuracy, precision, recall, F1 score, ROC-AUC, and/or balanced accuracy. Given that the primary concern is missing identification of at-risk students, recall and F1 scores would be particularly useful for comparison purposes among different models.

The project is achievable because the source dataset is public domain, structured and sufficiently small to process on a typical laptop or through Google Colab. Additionally, the methodology described above are also reasonable given the constraints of the course schedule since they are built upon established machine learning methodologies versus larger deep learning systems or custom infrastructures.

The project is relevant because there exists great pressure upon institutions of higher education to increase their ability to retain students and provide necessary support early in the student's educational experience; however, most interventions occur too late into the semester/term when a students' poor performance has already been identified. Therefore, if a functional early warning model existed that could assist advisors/administrators in determining what support should be provided first then many at-risk students might receive timely assistance.

Finally, the project is time-bound because the approach was taken with respect to the required capstone milestones from proposal/literature review to exploratory data analysis/modeling/validation/presentation. As previously stated, the intent was to limit the scope of the project so that it could be completed within one academic term.

Key Objectives

- Prepare and clean the selected student dataset for predictive modeling.
- Analyze the main patterns associated with dropout, continued enrollment, and graduation.
- Build and compare several machine learning classification models.
- Evaluate model performance using accuracy, precision, recall, F1-score, ROC-AUC, and balanced accuracy.
- Use SHAP-based explainability to identify the strongest predictors of dropout risk.
- Create a lightweight dashboard or prototype that presents predictions and explanations clearly.
- Translate the results into practical recommendations for early intervention in higher education.

Base Research Article

The source research article selected for this project was Islam et al. (2025) titled, "The integration of Explainable AI in Educational Data Mining for Student Academic Performance Prediction and Support System" (Islam et al., 2025). The selection of this article was appropriate for the capstone project because the article has been recently published, is a peer-reviewed journal publication, and addresses an issue (student risk prediction in higher education) that corresponds to the capstone's focus area. Additionally, the article utilizes the UCI Predict Students’ Dropout and Academic Success dataset as its data set, which aligns with the projected primary dataset for this capstone project.

This article presents a combination of machine learning techniques combined with explanatory artificial intelligence and decision-making thinking processes; both of these concepts are in line with the conceptualization for this capstone. As opposed to only presenting the accuracy of the predictive models, the authors include explanations (via SHAP and LIME) of why certain predictions were made. Therefore, based upon the criteria used to evaluate articles for use as a base paper versus simply using articles to establish a benchmark against which other articles could be evaluated, this article is considered a better base paper than a strictly technical benchmarking paper since it demonstrates how the predictive results can be utilized practically in an educational setting.

Therefore, there will be several differences in terms of comparison/extension from the article chosen as a base article. First, the focus of the capstone will be specifically on predicting students who are likely to drop out as early as possible (i.e. before dropping out occurs) and not on general student performance. Second, instead of evaluating only one or two baseline models, multiple baseline and advanced predictive models will be compared within the same workflow. Third, as opposed to focusing solely on metrics related to precision and classification accuracy, emphasis will be placed on recall, balanced accuracy, and identifying students who have the potential to drop out. Fourth, unlike the authors who chose to report their findings via reports on each model tested, findings will be presented via a simple dashboard-style prototype that resembles what a practitioner would view when using an advising tool.

3. Project Scope

3.1 Technical and Computing Constraints

The project is designed to stay within the technical and computing limits described in the course requirements. The primary dataset is well under 1 GB and can be processed on a normal personal laptop or in Google Colab without requiring GPU-based infrastructure. The planned workflow relies on standard Python libraries such as Pandas, Scikit-learn, XGBoost, and SHAP, all of which are practical to run in a local or free cloud environment.

In order to implement the project plans within a single academic term the implementation will utilize traditional tabular machine learning methods instead of more complex Deep Learning models or Enterprise Scale Systems. Also, the initial capstone stage of the project will NOT include Real-Time Deployment, Production APIs or Large-Scale Data Engineering. Again, these constraints have been intentionally put into place due to the fact that the desire is to create a robust and defendable Analytics Project vs. creating a Full Operational Platform..

3.2 Included Components

The project will include one main public dataset, a complete tabular machine learning workflow, and an explainability layer. The core models will be Logistic Regression, Decision Tree, Random Forest, and XGBoost. The analysis will include data cleaning, exploratory data analysis, feature engineering where needed, model training, cross-validation, and model comparison.

Feature importance and SHAP analysis will be used to explain why particular students are flagged as higher risk and which cohort-level variables matter most.

Python will be the main programming language, and the expected tools are Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, SHAP, and Jupyter Notebook or Google Colab. If time permits, the project will also include a simple Streamlit dashboard to display student outcome predictions, risk indicators, and model explanations in a more usable format.

3.3 Excluded Components

This project will be limited to a simulation of a real time deployment in an existing university information system; it will not utilize public facing Production API's, develop a mobile Application for use by students or faculty, nor will it integrate with an Enterprise System. Also this project will be limited in the use of private Institutional Data, Distributed Computing (with the exception of a single node) and Advanced Deep Learning Pipelines to ensure that the scope of this project can be completed as envisioned prior to expanding beyond the original Capstone Project goals.

4. Data Sourcing Plan

The primary dataset for this study will be the UCI Predict Students' Dropout and Academic Success. The primary dataset is being selected due to its alignment with the project's research question; having a clear definition of the outcome variable(s); and pre-existing structure that facilitates use for supervised learning.

Additionally, the number of records contained within the UCI dataset (which are estimated to be less than 10,000) make it practical to process under the computing limitations imposed by the course. Given these considerations, the UCI dataset is an attractive choice for Milestone 1 compared to other large and more complicated datasets available on education system. The primary rationale for selecting the UCI dataset over the OULAD as the primary dataset is based upon feasibility. While both are potentially useful sources for student academic success, OULAD provides additional information through behavioral logs. However, it is a more complicated dataset that requires processing multiple tables and temporal interactions. To minimize risks associated with the transition from a proposal to developing models and evaluating results during the time frame of a one term capstone project, using the cleaner UCI dataset is preferable to working first with the more complex OULAD.

Although OULAD is expected to provide a more comprehensive understanding of student learning behaviors when integrated into the framework developed above, it will serve as a backup or potential extension dataset. If early warnings generated via the proposed workflow do not perform effectively utilizing only the UCI dataset, then OULAD could be utilized to add strength to the project from a learning analytic perspective or test whether there are differences in performance when using a behavioral log version of the workflow.

5. Risk Assessment & Mitigation Plan

Beginning with the first possible risk - the selected data set does not include all factors important to dropout decisions in actuality; public data sets usually greatly oversimplify the true picture of an institution. In order to mitigate this risk, the OULAD model will be available for use as either a back-up or extension to the system being developed, and the final product will be framed as a decision support model as opposed to a comprehensive explanation of how students make choices.

The second potential risk is class imbalance. As long as there are fewer instances of dropouts (the class of interest) compared to non-dropout students (the other class), even though a model may have an impressive level of accuracy, it could very well perform badly when tested against those students who are at greatest risk. To help minimize this risk, the project team will not simply look at overall accuracy, but will consider the following metrics - Recall, F1-Score, ROC-AUC and Balanced Accuracy. The project team will also explore stratified sampling and potentially class weighting or SMOTE.

The third risk is overfitting, particularly for stronger modeling approaches like XGBoost. Overfitting can occur when a model performs extremely well on training data, but fails to generalize well to new examples. A variety of methods including cross-validation, comparing different models and conducting simple hyperparameter tuning will be used to address this concern. For this particular problem, the project team believes that simpler models will remain in scope during the remainder of the project. The reasons for keeping simpler models in scope includes both interpretability of the results and providing stable performance levels in addition to maximizing raw accuracy.

Finally, the last potential risk is that the project timeline and/or computing capabilities are overly constrained. Due to the fact that this project has to be completed prior to the end of one semester and that access to additional infrastructure is minimal, the technical approach outlined above was deliberately kept low-key: utilization of only one data set, testing of a limited number of models, and development of a light-weight user interface/dash board instead of full deployment.

6. References

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